

Assignment 4 :solns.

1. Find the functional forms for the spherical harmonics $Y_2^2(\Omega)$ and $Y_2^1(\Omega)$, By explicit integration, obtain the normalisation constants for them and show explicitly that they are orthogonal. What are the values of the corresponding energies and angular momenta?

$$\text{Ans. } Y_2^1(\Omega) = -N_1 \sin \theta. \cos \theta. e^{i\phi}$$

$$Y_2^2(\Omega) = -N_2 \sin^2 \theta. e^{2i\phi}$$

To find normalisation constants:

$$(N_1)^2 \int_0^\pi \sin^2 \theta. \cos^2 \theta. \sin \theta. d\theta \int_0^{2\pi} |e^{i\phi}|^2 d\phi = 1$$

$$\text{or, } (N_1)^2 \int_{-1}^1 (1-x^2) x^2. dx \int_0^{2\pi} d\phi = 1$$

$$\text{or, } (N_1)^2 \cdot \frac{4}{15} \cdot 2\pi = 1 \implies N_1 = \frac{1}{2} \sqrt{\frac{15}{2\pi}}$$

$$\text{and, } (N_2)^2 \int_0^\pi \sin^4 \theta. \sin \theta. d\theta \int_0^{2\pi} |e^{2i\phi}|^2 d\phi = 1$$

$$\text{or, } (N_2)^2 \int_{-1}^1 (1-x^2)^2. dx \int_0^{2\pi} d\phi = 1$$

$$\text{or, } (N_2)^2 \cdot \frac{16}{15} \cdot 2\pi = 1 \implies N_2 = \frac{1}{4} \sqrt{\frac{15}{2\pi}}$$

To prove orthogonality,

$$\text{we find: } \int_0^\pi \int_0^{2\pi} [Y_2^1(\Omega)]^* Y_2^2(\Omega). d\phi = \int_0^\pi \sin \theta. \cos \theta. \sin^2 \theta. \sin \theta. d\theta \int_0^{2\pi} e^{-i\phi}. e^{2i\phi}. d\phi$$

$$\int_0^\pi \sin^4 \theta. \cos \theta. d\theta \int_0^{2\pi} e^{i\phi}. d\phi = 0$$

Both integrals are separately zero.

The angular momentum depends only on the value of l .

$$\therefore \text{for each, angular momentum} = \sqrt{2(2+1)}\hbar = \sqrt{6}\hbar$$

The part about energy cannot be answered unless it is specified which system are we talking of. For example, if we are talking about a free particle on the surface of a sphere, then the energy = $\frac{L^2}{2I} = \frac{3\hbar^2}{I}$, where I =moment of inertia.

2. Using first order perturbation theory, calculate the energy of the n -th state for a particle of mass m moving in an infinite potential well of length $2L$ with wall at $x = 0$ and $x = 2L$, which is modified at the bottom by the perturbations:

(i) $\lambda V_0 \sin\left(\frac{\pi x}{2L}\right)$ and (ii) $\lambda V_0 \delta(x - L)$ where $\lambda \ll 1$.

Ans. The zeroth order solutions are:

$$\psi_n^{(0)}(\vec{r}) = \sqrt{\frac{1}{L}} \sin\left(\frac{n\pi}{2L}\right) x; \quad E_n^{(0)} = \frac{n^2 \hbar^2}{32mL^2}$$

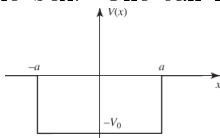
$$(i) H'_{nn} = \lambda V_0 \cdot \frac{1}{L} \int_0^{2L} \sin^2\left(\frac{n\pi x}{2L}\right) \sin\left(\frac{\pi x}{2L}\right) dx = \frac{\lambda V_0}{2L} \left[\int_0^{2L} \sin\left(\frac{\pi x}{2L}\right) dx - \int_0^{2L} \cos\left(\frac{n\pi x}{L}\right) \sin\left(\frac{\pi x}{2L}\right) dx \right]$$

$$\begin{aligned}
&= \frac{\lambda V_0}{2L} \left\{ \left[-\frac{2L}{\pi} \cos\left(\frac{\pi x}{2L}\right) \right]_0^{2L} - \frac{1}{2} \int_0^{2L} \left[\sin\left(\frac{(2n+1)\pi x}{2L}\right) - \sin\left(\frac{(2n-1)\pi x}{2L}\right) \right] dx \right\} \\
&= \frac{\lambda V_0}{2L} \left\{ \frac{4L}{\pi} - \frac{2L}{2\pi} \left[-\frac{1}{2n+1} \cos\left(\frac{(2n+1)\pi x}{2L}\right) + \frac{1}{2n-1} \cos\left(\frac{(2n-1)\pi x}{2L}\right) \right]_0^{2L} \right\} \\
&= \frac{\lambda V_0}{2L} \left\{ \frac{4L}{\pi} + \frac{L}{\pi} \left[\frac{2}{2n+1} - \frac{2}{2n-1} \right] \right\} = \frac{\lambda V_0}{\pi} \left(2 + \frac{1}{2n+1} - \frac{1}{2n-1} \right) = \frac{\lambda V_0}{\pi} \frac{8n^2}{1-4n^2}
\end{aligned}$$

$$(ii) \quad E^{(1)} = H'_{nn} = \frac{\lambda V_0}{L} \int_0^{2L} \delta(x-L) \sin^2\left(\frac{n\pi}{2L}x\right) dx = \frac{\lambda V_0}{L} \sin^2\left(\frac{n\pi}{2}\right) = \begin{cases} \frac{\lambda V_0}{L} \quad \forall n \text{ odd} \\ 0 \quad \forall n \text{ even} \end{cases}; \quad \therefore E = E^{(0)} + E^{(1)} =$$

$$\frac{n^2 h^2}{32mL^2} + \begin{cases} \frac{\lambda V_0}{L} \quad \forall n \text{ odd} \\ 0 \quad \forall n \text{ even} \end{cases}$$

3. The particle in a box problem is also called an infinite well problem, since the potential energy at the walls goes to infinity and it is zero inside the box. One can instead construct a 'finite' square well problem with P.E. :

$$V(x) = \begin{cases} -V_0 & \forall -a \leq x \leq a \\ 0 & \forall |x| > a \end{cases} \quad \text{with } V_0 > 0.$$


(a) For a particle of mass m and energy $E < 0$ (bound states), with $\hbar\kappa = \sqrt{-2mE}$ and $\hbar l = \sqrt{2m(E + V_0)}$, show that $\kappa = l \tan(la)$.

Use $z = la$ and $\hbar z_0 = a\sqrt{2mV_0}$ and obtain $\tan z = \sqrt{\left(\frac{z_0}{z}\right)^2 - 1}$.

Plot $\tan z$ and the RHS as functions of z and discuss solutions for the following cases :

1. Wide, deep well (and the limit $V_0 \rightarrow \infty$); 2. Shallow, normal well.

(b) For a particle with energy $E > 0$ (scattering states), find the transmission coefficient and discuss how it varies with the energy of the particle.

Ans. See p.70-74 of the text by Griffiths and Schroeter,. attached separately.

4. For the Hamiltonian, with $\lambda \ll 1$,

$$E_0 \begin{pmatrix} 1 + \lambda & 0 & 0 & 0 \\ 0 & 8 & 0 & 0 \\ 0 & 0 & 3 & -2\lambda \\ 0 & 0 & -2\lambda & 7 \end{pmatrix}$$

(a) find the eigenvalues and eigenvectors of the unperturbed Hamiltonian ($\lambda = 0$), (b) diagonalize the full Hamiltonian to find the exact eigenvalues and expand each eigenvalue to the second power of λ and (c) using the first and second order perturbation theory find the approximate energy eigenvalues and eigenstates of the full Hamiltonian.

Ans.

$$H_0 = E_0 \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 8 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 7 \end{pmatrix}; \quad H' = E_0 \begin{pmatrix} \lambda & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2\lambda \\ 0 & 0 & -2\lambda & 0 \end{pmatrix}$$

(a) The eigenvectors for the diagonal matrix are, $|1\rangle \equiv \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$; $|2\rangle \equiv \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$; $|3\rangle \equiv \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$; $|4\rangle \equiv \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$; the

corresponding eigenvalues are 1, 8, 3, 7 respectively.

(b) To diagonalise the full Hamiltonian, we need to find the eigenvalues of the matrix $E_0 \begin{pmatrix} 1+\lambda & 0 & 0 & 0 \\ 0 & 8 & 0 & 0 \\ 0 & 0 & 3 & -2\lambda \\ 0 & 0 & -2\lambda & 7 \end{pmatrix}$

These are, the solutions of the quartic equation: $\det \begin{vmatrix} 1+\lambda-a & 0 & 0 & 0 \\ 0 & 8-a & 0 & 0 \\ 0 & 0 & 3-a & -2\lambda \\ 0 & 0 & -2\lambda & 7-a \end{vmatrix} = 0$

or, $E_1 = E_0(1+\lambda)$; $E_2 = 8E_0$ and $E_3 = a_3 E_0$; $E_4 = a_4 E_0$ where a_3 and a_4 are solutions of $\det \begin{vmatrix} 3-a & -2\lambda \\ -2\lambda & 7-a \end{vmatrix} = 0$

or, $a^2 - 10a + (21 - 4\lambda^2) = 0 \implies a_{3(4)} = \frac{10 \pm \sqrt{100 - 4(21 - 4\lambda^2)}}{2} = 5 \pm 2\sqrt{1 + \lambda^2} \approx \left[5 \pm 2 \left(1 + \frac{\lambda^2}{2} \right) \right] = 7 + \lambda^2$, or,

$3 - \lambda^2$

$\therefore$ up to second order exact $E_{3(4)} = E_0(3 - \lambda^2)$ and $E_0(7 + \lambda^2)$.

(c) The zeroth order energies are: $E_1^{(0)} = E_0$; $E_2^{(0)} = 8E_0$; $E_3^{(0)} = 3E_0$; $E_4^{(0)} = 7E_0$ (they are not in order)

for states, $|1\rangle \equiv \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$; $|2\rangle \equiv \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$; $|3\rangle \equiv \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$; $|4\rangle \equiv \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$; respectively ;

The corrections upto second order :

In the diagonal basis of eigenvectors, the matrix elements can be directly picked up from the H' matrix, i.e.,

$H'_{11} = \lambda E_0$; $H'_{22} = H'_{33} = H'_{44} = 0$; $H'_{12} = H'_{21} = H'_{31} = H'_{13} = H'_{41} = H'_{14} = H'_{23} = H'_{32} = H'_{24} = H'_{42} = 0$;

and $H'_{34} = H'_{43} = -2\lambda E_0$:

$$E_1^{(1)} = H'_{11} = \lambda E_0;$$

$$E_1^{(2)} = \frac{|H'_{12}|^2}{E_0 - 8E_0} + \frac{|H'_{13}|^2}{E_0 - 3E_0} + \frac{|H'_{14}|^2}{E_0 - 7E_0} = 0$$

Hence, upto second order, corrected energy = $E_0 (1 + \lambda)$, same as the exact solution for energy.

$$E_2^{(1)} = H'_{22} = 0$$

$$E_2^{(2)} = \frac{|H'_{12}|^2}{8E_0 - E_0} + \frac{|H'_{23}|^2}{8E_0 - 3E_0} + \frac{|H'_{24}|^2}{8E_0 - 7E_0} = 0$$

Hence, upto second order, corrected energy = $8E_0$, same as the exact solution for energy.

$$E_3^{(1)} = H'_{33} = 0$$

$$E_3^{(2)} = \frac{|H'_{13}|^2}{3E_0 - E_0} + \frac{|H'_{23}|^2}{3E_0 - 8E_0} + \frac{|H'_{34}|^2}{3E_0 - 7E_0} = -\lambda^2 E_0$$

Hence, upto second order, corrected energy = $E_0 (3 - \lambda^2)$, same as the exact solution for energy.

$$E_4^{(1)} = H'_{44} = 0$$

$$E_4^{(2)} = \frac{|H'_{14}|^2}{7E_0 - E_0} + \frac{|H'_{24}|^2}{7E_0 - 8E_0} + \frac{|H'_{34}|^2}{7E_0 - 3E_0} = \lambda^2 E_0$$

Hence, upto second order, corrected energy = $E_0 (7 + \lambda^2)$, same as the exact solution for energy.

The corrected states (upto first order) are as follows :

$$|\psi_1\rangle = N_1(|1\rangle + \frac{H'_{21}}{E_1^0 - E_2^0}|2\rangle + \frac{H'_{31}}{E_1^0 - E_3^0}|3\rangle + \frac{H'_{41}}{E_1^0 - E_4^0}|4\rangle) = N_1(|1\rangle) = |1\rangle.$$

$$|\psi_2\rangle = N_2(\frac{H'_{12}}{E_2^0 - E_1^0}|1\rangle + |2\rangle + \frac{H'_{32}}{E_2^0 - E_3^0}|3\rangle + \frac{H'_{42}}{E_2^0 - E_4^0}|4\rangle) = N_2(|2\rangle) = |2\rangle$$

$$|\psi_3\rangle = N_3(\frac{H'_{13}}{E_3^0 - E_1^0}|1\rangle + \frac{H'_{23}}{E_3^0 - E_2^0}|2\rangle + |3\rangle + \frac{H'_{43}}{E_3^0 - E_4^0}|4\rangle) = N_3(|3\rangle + \frac{2\lambda}{4\sqrt{1+\lambda^2}}|4\rangle) \approx N_3(|3\rangle + \frac{\lambda}{2}|4\rangle); N_3 = \frac{1}{\sqrt{1+\frac{\lambda^2}{4}}}$$

$$|\psi_4\rangle = N_4(\dots + \frac{H'_{34}}{E_4^0 - E_3^0}|3\rangle + |4\rangle) = N_4(\frac{2\lambda}{4\sqrt{1+\lambda^2}}|3\rangle + |4\rangle) \approx N_4 - \frac{\lambda}{2}(|3\rangle + \frac{\lambda}{2}|4\rangle); N_4 = \frac{1}{\sqrt{1+\frac{\lambda^2}{4}}}$$

5. Let $\vec{A}$ be a vector operator s.t. $[L_i, A_j] = i\hbar\epsilon_{ijk}L_k$

$$\text{Show: (i) } \vec{L} \cdot \vec{A} = \vec{A} \cdot \vec{L} \text{ (ii) } [L_i, \vec{L} \cdot \vec{A}] = [L^2, \vec{L} \cdot \vec{A}] = [L^2, L_i \vec{L} \cdot \vec{A}] = 0$$

$$\text{(iii) } [L^2, [L^2, A_i]] = 2\hbar^2 \{L^2 A_i + A_i L^2 - 2L_i \vec{L} \cdot \vec{A}\} \text{ [use: } \epsilon_{ijk}\epsilon_{lmk} = \delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}]$$

$$\text{Ans. (i) } \because [L_1, A_1] = [L_2, A_2] = [L_3, A_3] = 0, \therefore \vec{L} \cdot \vec{A} = \vec{A} \cdot \vec{L}$$

$$\text{(ii) } [L_i, \vec{L} \cdot \vec{A}] = [L_i, L_j A_j] = [L_i, L_j] A_j + L_j [L_i, A_j] = i\hbar\epsilon_{ijk} \{L_k A_j + L_j A_k\} = 0$$

To see the last part of the above, note that ϵ_{ijk} is asymmetric and $\{L_k A_j + L_j A_k\}$ is symmetric under $j \leftrightarrow k$ reversal.

$$[L^2, \vec{L} \cdot \vec{A}] = L_i [L_i, \vec{L} \cdot \vec{A}] + [L_i, \vec{L} \cdot \vec{A}] L_i = 0$$

$$[L^2, L_i \vec{L} \cdot \vec{A}] = L_i [L^2, \vec{L} \cdot \vec{A}] + [L^2, L_i] \vec{L} \cdot \vec{A} = 0$$